

The record estimate H_{record} :
exact structure, equivalence with the Riemann Hypothesis,
and the obstruction in the prime-side expression

Christophe Michaels

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Answer

Of the four responses requested, the one that can be given is the fourth.

1. No proof of H_{record} is given.
2. No theorem in the supplied research implies H_{record} . Theorem 3.1 shows that H_{record} is equivalent to the Riemann Hypothesis, and the atlas entry M38, which is where the debit identity and the record induction come from, already labels the debit estimate “open and equivalent in strength to the target”. The block, cofactor-flow and covariance results are exact identities or bounds in terms of the heights of M ; Section 5 checks each one’s definitions and scope and finds that each transfers correctly and moves the whole difficulty into those heights.
3. No weaker sufficient alternative exists. The closing deduction ends in $R(N) \ll_{\varepsilon} N^{\varepsilon}$, which is itself equivalent to the Riemann Hypothesis (Theorem 3.1), so every statement that closes the same argument has the strength of the Riemann Hypothesis.
4. What is established here, with proofs: the exact structure of the debit in terms of the smooth Möbius sum $A(k)$ and the high-prime sum $S(k)$, in which the two enter only through $A - S = M$ (Proposition 2.4); the exponent dictionary, in which H_{record} with a fixed exponent θ is a zero-free half-plane $\Re s > \frac{1}{2} + \theta$ and the unconditional exponent is $1 - o(1)$ (Theorem 3.2); the lower bounds $E(W) \geq (1 - o(1))m/(24 \log^2 m)$ and $E(u_0) + 2M(m)h(u_0) \geq (1 - o(1))m/(24 \log^2 m)$, so that the signed cross term $2\langle a, W \rangle$ is positive and cancels both to within Δ_N (Theorem 4.1); the bound $\sum_p E(v_p) \leq (\log 2 + o(1))R^*(m)$ for the diagonal of the prime-copy Gram matrix, so that the off-diagonal covariance carries all of $E(W)$ but at most $(\log 2 + o(1))R^*(m)$ (Corollary 4.2); and the exact statement of the unsupported step (Section 6).

Status labels. Following the fixed dictionary, every statement is one of: **Definition**; **Proved** (here, or cited with its hypotheses); **Remaining hypothesis**. Numerical values appear only in Section 7, as verification of identities and of orders of magnitude, and are labelled **Computation**.

1 Definitions

Definition 1.1 (Energy). For finitely supported real coefficient vectors f, g on the positive integers,

$$\langle f, g \rangle = \sum_{i, j \geq 1} \frac{f(i)g(j)}{\max(i, j)}, \quad E(f) = \langle f, f \rangle.$$

The sum does not depend on the length N of the ambient space, so $\langle f, g \rangle_N = \langle f, g \rangle$ and $E_N(f) = E(f)$ whenever $\text{supp } f, \text{supp } g \subseteq [1, N]$. With μ the Möbius function and $M(x) = \sum_{n \leq x} \mu(n)$,

$$R(N) = E(\mu \mathbf{1}_{[1, N]}), \quad R(0) = 0.$$

$N \geq 1$ is a *strict energy record* when $R(N) > \max_{0 \leq n < N} R(n)$. For $N \geq 1$ put $m = \lfloor \sqrt{N} \rfloor$ and $\Delta_N = R(N) - R(m)$.

Definition 1.2 (The target and its fixed-exponent version). H_{record} is the statement: for every $\eta > 0$ there are $A_\eta < \infty$ and N_η such that $\Delta_N \leq A_\eta N^\eta R(m)$ for every strict energy record $N \geq N_\eta$. For a fixed $\theta \geq 0$, $H_{\text{record}}(\theta)$ is the statement: there are $A < \infty$ and N_0 with $\Delta_N \leq AN^\theta R(m)$ at every strict energy record $N \geq N_0$. Thus H_{record} is $H_{\text{record}}(\theta)$ for every $\theta > 0$.

Definition 1.3 (Prime-side objects). Let $P^+(n)$ be the largest prime factor of n , $P^+(1) = 1$, and let e_j be the unit vector at j . For $N \geq 1$ and $m = \lfloor \sqrt{N} \rfloor$:

$$a = \mu \mathbf{1}_{\{P^+ \leq m\}} \mathbf{1}_{[1, N]}, \quad u_0 = a \mathbf{1}_{(m, N]}, \quad v_p = \sum_{k \leq N/p} \mu(k) e_{pk}, \quad W = \sum_{m < p \leq N} v_p,$$

p running over primes, and $h(f) = \sum_{m < n \leq N} f(n)/n$ for f supported in $(m, N]$. Write U_p for the dilation $U_p e_k = e_{pk}$, $L_p = \lfloor N/p \rfloor$, so that $v_p = U_p(\mu \mathbf{1}_{[1, L_p]})$, and

$$A(k) = \sum_{n \leq k} a(n), \quad S(k) = \sum_{n \leq k} W(n), \quad m_1(x) = \sum_{n \leq x} \frac{\mu(n)}{n}, \quad R^*(m) = \max_{1 \leq n \leq m} R(n).$$

Every integer $n \leq N$ has at most one prime factor exceeding m , since $(m+1)^2 > N$; if it has one, p , then $n = pk$ with $k \leq N/p < m+1$, so $k \leq m$ and $\mu(n) = -\mu(k)$. Hence $\mu \mathbf{1}_{[1, N]} = a - W$ and $a = \mu \mathbf{1}_{[1, m]} + u_0$.

2 Exact identities

All statements in this section are **Proved**.

Lemma 2.1 (Cumulative form). Let f, g be supported in $[1, N]$ with partial sums $F(k) = \sum_{n \leq k} f(n)$, $G(k) = \sum_{n \leq k} g(n)$. Then

$$\langle f, g \rangle = \sum_{k=1}^{N-1} \frac{F(k)G(k)}{k(k+1)} + \frac{F(N)G(N)}{N}.$$

In particular $R(N) = \sum_{k < N} M(k)^2/(k(k+1)) + M(N)^2/N$, $R(N) \geq M(N)^2/N$, and $R(N) \geq \frac{1}{2}$ for every $N \geq 1$.

Proof. $1/\max(i, j) = \sum_{k \geq \max(i, j)} (\frac{1}{k} - \frac{1}{k+1})$. Insert this in $\langle f, g \rangle$ and interchange the finite sums: the coefficient of $\frac{1}{k} - \frac{1}{k+1}$ is $\sum_{i, j \leq \min(k, N)} f(i)g(j) = F(\min(k, N))G(\min(k, N))$, and $\sum_{k \geq N} (\frac{1}{k} - \frac{1}{k+1}) = 1/N$. For the last claim, $R(1) = 1$ and for $N \geq 2$ the term $k = 1$ equals $M(1)^2/2 = \frac{1}{2}$. $\square$

Lemma 2.2 (The history enters through two scalars). *If g is supported in $(m, N]$ then $\langle \mu \mathbf{1}_{[1,m]}, g \rangle = M(m)h(g)$, and $E(\mu \mathbf{1}_{[1,m]}) = R(m)$.*

Proof. For $i \leq m < j$, $\max(i, j) = j$, so $\langle \mu \mathbf{1}_{[1,m]}, g \rangle = \sum_{j>m} \frac{g(j)}{j} \sum_{i \leq m} \mu(i)$. The second claim is the N -independence in Definition 1.1. $\square$

Proposition 2.3 (The debit identity). *For every $N \geq 1$,*

$$\Delta_N = E(u_0) + 2M(m)h(u_0) + E(W) - 2\langle a, W \rangle \quad (1)$$

and

$$\Delta_N = \sum_{k=m}^{N-1} \frac{M(k)^2}{k(k+1)} + \frac{M(N)^2}{N} - \frac{M(m)^2}{m}. \quad (2)$$

Proof. Write $\mu \mathbf{1}_{[1,N]} = \mu \mathbf{1}_{[1,m]} + (u_0 - W)$ with $u_0 - W$ supported in $(m, N]$. Expanding and using Lemma 2.2,

$$\begin{aligned} R(N) &= R(m) + E(u_0 - W) + 2M(m)(h(u_0) - h(W)) \\ &= R(m) + E(u_0) + E(W) - 2\langle u_0, W \rangle + 2M(m)h(u_0) - 2M(m)h(W). \end{aligned}$$

Since $a = \mu \mathbf{1}_{[1,m]} + u_0$, Lemma 2.2 gives $\langle a, W \rangle = M(m)h(W) + \langle u_0, W \rangle$, which is (1). For (2), subtract the cumulative forms of $R(N)$ and $R(m)$ and use $\sum_{k=m}^{N-1} \frac{1}{k(k+1)} + \frac{1}{N} = \frac{1}{m}$ for the terminal term $M(m)^2/m$ of $R(m)$. $\square$

Proposition 2.4 (Partial sums of the prime-side vectors). *(i) $A(k) = M(k)$ for $k \leq m$, and $A(k) = M(k) + S(k)$ for all $k \leq N$.*

(ii) $S(k) = \sum_{m < p \leq k} M(\lfloor k/p \rfloor) = \sum_{j \leq k/(m+1)} \mu(j)(\pi(k/j) - \pi(m))$ for $k \leq N$.

(iii) $S(k) = 0$ for $k \leq m$, and $S(k) = \pi(k) - \pi(m)$ for $m < k < 2(m+1)$.

(iv) The three components of (1) are

$$E(u_0) + 2M(m)h(u_0) = \sum_{k=m}^{N-1} \frac{A(k)^2}{k(k+1)} + \frac{A(N)^2}{N} - \frac{M(m)^2}{m}, \quad (3)$$

$$E(W) = \sum_{k=m+1}^{N-1} \frac{S(k)^2}{k(k+1)} + \frac{S(N)^2}{N}, \quad (4)$$

$$\langle a, W \rangle = \sum_{k=m+1}^{N-1} \frac{A(k)S(k)}{k(k+1)} + \frac{A(N)S(N)}{N}. \quad (5)$$

(v) Termwise, $A(k)^2 + S(k)^2 - 2A(k)S(k) = (A(k) - S(k))^2 = M(k)^2$: the regrouping (1) is an identity in which A and S enter only through their difference.

Proof. (i) Every $n \leq m$ is m -smooth, so $a = \mu$ on $[1, m]$; and $a = \mu \mathbf{1}_{[1,N]} + W$ gives $A = M + S$. (ii) $W(n) = \mu(n/p)$ when n has the (unique) prime factor $p > m$ and $W(n) = 0$ otherwise, so $S(k) = \sum_{m < p \leq k} \sum_{j \leq k/p} \mu(j)$; grouping by j counts the primes p with $m < p \leq k/j$. (iii) If $m < n \leq k < 2(m+1)$ and $n = pj$ with $p > m$, then $j < 2$, so $n = p$ and $W(n) = \mu(1) = 1$.

(iv) u_0 has partial sums $A(k) - M(m)$ on $[m, N]$ and 0 before; Lemma 2.1 and Abel summation $h(u_0) = \sum_{k=m+1}^{N-1} \frac{A(k)-M(m)}{k(k+1)} + \frac{A(N)-M(m)}{N}$ give

$$E(u_0) + 2M(m)h(u_0) = \sum_{k=m+1}^{N-1} \frac{A(k)^2 - M(m)^2}{k(k+1)} + \frac{A(N)^2 - M(m)^2}{N},$$

and $\sum_{k=m+1}^{N-1} \frac{1}{k(k+1)} + \frac{1}{N} = \frac{1}{m+1}$, $\frac{1}{m} - \frac{1}{m+1} = \frac{1}{m(m+1)}$, $A(m) = M(m)$ give (3). (4) and (5) are Lemma 2.1 with $S(k) = 0$ for $k \leq m$. (v) is algebra, and inserting it in (3)–(5) recovers (2). $\square$

Remark 2.5. The identity (1) is the complete debit D_N of atlas entry M38 and the two-scalar update $R(N) = R(m) + F + 2Ah$ of entry M37. Proposition 2.4(v) is the precise sense in which the regrouping “retains the inherited history, every prime-copy cross term and the terminal”: it retains them because it is an identity, and for the same reason it cannot by itself make any of them smaller.

3 Equivalence with the Riemann Hypothesis

Theorem 3.1 (Proved). *The following are equivalent.*

- (i) *The Riemann Hypothesis.*
- (ii) $R(N) \ll_{\varepsilon} N^{\varepsilon}$ for every $\varepsilon > 0$.
- (iii) For every $\eta > 0$ there is A_{η} with $\Delta_N \leq A_{\eta} N^{\eta} R(m)$ for every $N \geq 1$.
- (iv) H_{record} .

Proof. (i) $\Leftrightarrow$ (ii) is the theorem “The Green energy and the Riemann Hypothesis” of [1, §12]; for completeness: (i) gives $M(x) \ll x^{1/2+\varepsilon/2}$ [3, §14.25], hence $R(N) \leq \sum_{k \leq N} M(k)^2/k^2 \ll \sum_{k \leq N} k^{\varepsilon-1} \ll N^{\varepsilon}$; conversely, if $R(N) \leq CN^{\varepsilon}$, then by Lemma 2.1 and Cauchy–Schwarz on dyadic blocks, for $\sigma > \frac{1}{2}$,

$$\begin{aligned} \sum_{K < k \leq 2K} \frac{|M(k)|}{k^{1+\sigma}} &\leq K^{-\sigma} \left(\sum_{K < k \leq 2K} 1 \right)^{1/2} \left(\sum_{K < k \leq 2K} \frac{M(k)^2}{k^2} \right)^{1/2} \\ &\leq K^{\frac{1}{2}-\sigma} (2C(2K+1)^{\varepsilon})^{1/2}, \end{aligned}$$

which is summable over $K = 2^j$ when $\sigma > \frac{1}{2} + \frac{\varepsilon}{2}$; so $\sum_n \mu(n)n^{-s} = s \int_1^{\infty} M(x)x^{-s-1}dx$ converges absolutely for $\Re s > \frac{1}{2} + \frac{\varepsilon}{2}$ and represents $1/\zeta(s)$ there by analytic continuation from $\Re s > 1$. Thus ζ has no zeros in $\Re s > \frac{1}{2}$, and by the functional equation none in $\Re s < \frac{1}{2}$.

(ii) $\Rightarrow$ (iii): $\Delta_N \leq R(N) \leq C_{\eta} N^{\eta} \leq 2C_{\eta} N^{\eta} R(m)$ by Lemma 2.1, since $R(m) \geq \frac{1}{2}$ for $m \geq 1$. (iii) $\Rightarrow$ (iv) is a restriction to records.

(iv) $\Rightarrow$ (ii) (the record induction of M38). Fix $\varepsilon > 0$, put $\eta = \varepsilon/4$, take A_{η}, N_{η} from H_{record} , and choose $N_1 \geq N_{\eta}$ with $N_1^{\varepsilon/4} \geq 1 + A_{\eta}$. Let $C = \max\{1, \max_{n \leq N_1} R(n)\}$. We show $R(n) \leq Cn^{\varepsilon}$ for all $n \geq 1$ by strong induction. It holds for $n \leq N_1$. Let $N > N_1$. If N is not a strict record, $R(N) \leq \max_{n < N} R(n) \leq C \max_{n < N} n^{\varepsilon} \leq CN^{\varepsilon}$. If N is a strict record, with $m = \lfloor \sqrt{N} \rfloor < N$,

$$R(N) = R(m) + \Delta_N \leq (1 + A_{\eta} N^{\varepsilon/4})R(m) \leq (1 + A_{\eta})N^{\varepsilon/4} C m^{\varepsilon} \leq (1 + A_{\eta})C N^{\varepsilon/4+\varepsilon/2} \leq CN^{\varepsilon}. \quad \square$$

Theorem 3.2 (Exponent dictionary; **Proved**). *Let $0 \leq \theta < 1$.*

- (a) If $H_{\text{record}}(\theta)$ holds, then $R(N) \ll_{\delta} N^{2\theta+\delta}$ for every $\delta > 0$, and $\zeta(s) \neq 0$ for $\Re s > \frac{1}{2} + \theta$.
- (b) If $\zeta(s) \neq 0$ for $\Re s > \frac{1}{2} + \theta$, then for every $\delta > 0$ there is A with $\Delta_N \leq AN^{2\theta+\delta}R(m)$ for all $N \geq 1$; in particular $H_{\text{record}}(2\theta + \delta)$ holds.
- (c) Unconditionally there is $c > 0$ with $\Delta_N \leq 2N \exp(-c(\log N)^{3/5}(\log \log N)^{-1/5})R(m)$ for all $N \geq 3$. No $\theta < 1$ for which $H_{\text{record}}(\theta)$ holds is known.

Proof. (a) Run the induction of Theorem 3.1 with the exponent $\varphi = 2\theta + \delta$: at a record, $R(N) \leq (1 + AN^{\theta})Cm^{\varphi} \leq (1 + A)CN^{\theta+\theta+\delta/2} \leq CN^{\varphi}$ once $N^{\delta/2} \geq 1 + A$. The dyadic Cauchy–Schwarz bound then gives $K^{\frac{1}{2}-\sigma}(2C(2K+1)^{\varphi})^{1/2}$, summable for $\sigma > \frac{1}{2} + \theta + \frac{\delta}{2}$; so $1/\zeta$ is holomorphic in $\Re s > \frac{1}{2} + \theta + \frac{\delta}{2}$ for every $\delta > 0$. (b) The absence of zeros in $\Re s > \frac{1}{2} + \theta$ gives $M(x) \ll_{\delta} x^{1/2+\theta+\delta/2}$; the proofs of Theorems 14.2 and 14.25 of [3] use the zero-free half-plane only to make $\log \zeta$ holomorphic in the discs to which the Borel–Carathéodory lemma is applied, and apply verbatim with $\frac{1}{2} + \theta$ in place of $\frac{1}{2}$. Then $R(N) \leq \sum_{k \leq N} M(k)^2/k^2 \ll N^{2\theta+\delta}$ and $\Delta_N \leq R(N) \leq 2R(N)R(m)$. (c) $M(x) \ll x \exp(-c_0(\log x)^{3/5}(\log \log x)^{-1/5})$ [4, Theorem 12.7]; summing $M(k)^2/k^2$ over $k \leq N$, the terms with $k \leq N \exp(-(\log N)^{3/5})$ contribute at most that bound, and the others at most $N \exp(-2c_0(1 - o(1))(\log N)^{3/5}(\log \log N)^{-1/5})$. Finally $\Delta_N \leq R(N) \leq 2R(N)R(m)$. $\square$

Remark 3.3 (What the dictionary says about the task). The target asks for $H_{\text{record}}(\theta)$ for every $\theta > 0$. By (a), a proof for any single $\theta < \frac{1}{2}$ would already give a zero-free half-plane of positive width, which is not known; and by (c) the exponent that is known is $1 - o(1)$. Every intermediate improvement is a zero-free half-plane and the full target is the Riemann Hypothesis. The induction of M38 is correct and is reproduced above; it is the estimate it consumes that has this strength.

4 The size of the components

Theorem 4.1 (Proved). As $N \rightarrow \infty$, with $m = \lfloor \sqrt{N} \rfloor$,

$$E(W) \geq (1 - o(1)) \frac{m}{24 \log^2 m}, \quad (6)$$

$$E(u_0) + 2M(m)h(u_0) \geq (1 - o(1)) \frac{m}{24 \log^2 m}, \quad (7)$$

$$\langle a, W \rangle \geq (1 - o(1)) \frac{m}{24 \log^2 m} - \frac{\Delta_N}{2}. \quad (8)$$

In particular, whenever $\Delta_N = o(m/\log^2 m)$, the cross term $2\langle a, W \rangle$ is positive and equals $E(u_0) + 2M(m)h(u_0) + E(W)$ to within Δ_N : the debit is the difference of quantities each of size at least $m/(25 \log^2 m)$.

Proof. Let $m \geq 3$, so that $2m \leq N - 1$. Put $k_0 = \lceil 3m/2 \rceil$. By Proposition 2.4(iii), $S(k) = \pi(k) - \pi(m) \geq \pi(k_0) - \pi(m)$ for $k_0 \leq k \leq 2m$. Dropping the other nonnegative terms of (4),

$$E(W) \geq (\pi(k_0) - \pi(m))^2 \sum_{k=k_0}^{2m} \frac{1}{k(k+1)} = (\pi(k_0) - \pi(m))^2 \left(\frac{1}{k_0} - \frac{1}{2m+1} \right).$$

By the prime number theorem $\pi(k_0) - \pi(m) = (1 + o(1)) \frac{m}{2 \log m}$, and $\frac{1}{k_0} - \frac{1}{2m+1} = (1 + o(1)) \frac{1}{6m}$; this is (6).

For (7), drop the nonnegative terms of (3) outside $[k_0, 2m]$ and use $A(k) = M(k) + \pi(k) - \pi(m)$ there. Since $M(x) \ll x \exp(-c\sqrt{\log x})$ (the prime number theorem with the de la Vallée Poussin

error term [4, Ch. 12]), $\max_{k \leq 2m} |M(k)| = o(m/\log m)$, so $A(k) \geq (1 - o(1)) \frac{m}{2 \log m}$ on $[k_0, 2m]$, and $M(m)^2/m = o(m/\log^2 m)$. The same telescoping sum gives (7). Finally (8) is (1) solved for $\langle a, W \rangle$. $\square$

Corollary 4.2 (The diagonal of the prime-copy Gram matrix; **Proved**). $E(v_p) = R(L_p)/p$, and

$$\begin{aligned} \sum_{m < p \leq N} E(v_p) &= \sum_{m < p \leq N} \frac{R(\lfloor N/p \rfloor)}{p} \leq (\log 2 + o(1)) R^*(m), \\ \sum_{\substack{m < p, q \leq N \\ p \neq q}} \langle v_p, v_q \rangle &\geq (1 - o(1)) \frac{m}{24 \log^2 m} - (\log 2 + o(1)) R^*(m). \end{aligned}$$

Proof. $E(U_p f) = E(f)/p$ because $\max(pi, pj) = p \max(i, j)$. Since $1 \leq L_p \leq m$, $R(L_p) \leq R^*(m)$, and $\sum_{m < p \leq N} 1/p = \log \frac{\log N}{\log m} + O(1/\log m) = \log 2 + o(1)$ by Mertens' theorem. The second display is $E(W) = \sum_{p, q} \langle v_p, v_q \rangle$ with (6). $\square$

Corollary 4.3 (No term-by-term proof in the regime of the target; **Proved**). Suppose $R(n) \leq Cn^{\varepsilon_0}$ for all $n \geq 1$, with some $\varepsilon_0 < 1$. Then for all large N ,

$$E(W) \geq \frac{m^{1-\varepsilon_0}}{25C \log^2 m} R(m) \quad \text{and} \quad E(u_0) + 2M(m)h(u_0) \geq \frac{m^{1-\varepsilon_0}}{25C \log^2 m} R(m).$$

Hence, in the regime $R(N) \ll N^\varepsilon$ that the closing deduction is meant to reach, no inequality of the form $E(W) \leq A_\eta N^\eta R(m)$ or $E(u_0) + 2M(m)h(u_0) \leq A_\eta N^\eta R(m)$ can hold with $\eta < \frac{1-\varepsilon}{2}$. A proof of H_{record} can only be a bound on the signed combination (1), to relative precision $N^{\eta-1/2}$ in its terms.

Proof. Divide (6) and (7) by $R(m) \leq Cm^{\varepsilon_0}$, and use $m \geq \sqrt{N} - 1$. $\square$

Remark 4.4 (The size of S beyond $2m$; not proved here). For $k \in (m, 2m)$ the high-prime sum is $\pi(k) - \pi(m)$ by Proposition 2.4(iii). For k of the order of N , formal expansion of $\sum_{j \leq k/(m+1)} \mu(j) \text{li}(k/j)$ with $\sum_j \mu(j)/j = 0$ and $\sum_j \mu(j) \log j/j = -1$ suggests $S(k) \sim -k/\log^2 k$ and $E(W) \asymp N/\log^4 N$; the verification in Section 7 finds $S(k) \log^2 k/k$ between -1.0 and -1.3 for $N/8 \leq k \leq N$ at $N = 10^6$, and $E(W)/(N/\log^4 N) \approx 3.7$ at $N = 10^6$ and 10^7 . This stronger lower bound is not needed for Corollary 4.3 and is not claimed as proved.

5 The supplied results, checked against the components

Each supplied result is stated with its hypotheses, verified, and then matched against (1). The conclusion in every case is the same: the result is exact or one-sided in the heights of M , it transfers correctly, and after the transfer the quantity to be bounded is again the heights of M over $(m, N]$.

5.1 Completed and unfinished blocks

Proved [1, Theorems on future orthogonality, block energy and the native block budget, §§5, 8, 11]. For a block $B = (a, b]$ on which $M(k) - M(a)$ returns to 0 at $k = b$, the restriction $f_B = \mu \mathbf{1}_B$ has zero mass, is orthogonal to every vector supported in (b, ∞) , and $E(f_B) = \sum_{k=a+1}^{b-1} (M(k) - M(a))^2/(k(k+1)) \leq H_B^2 \left(\frac{1}{a+1} - \frac{1}{b} \right)$ with $H_B = \max_{a < k < b} |M(k) - M(a)|$. An unfinished block $(a, N]$ has energy at most $H_{\text{open}}^2/(a+1)$.

Transfer. Apply the first-return rule to $\mu \mathbf{1}_{(m,N]}$ from $a_0 = m$. With $F = E(\mu \mathbf{1}_{(m,N]}) = \sum_B E(f_B) + E_{\text{open}}$ and Lemma 2.2,

$$\Delta_N = F + 2M(m)h(\mu \mathbf{1}_{(m,N]}), \quad F \leq \sum_B H_B^2 \left(\frac{1}{a_B + 1} - \frac{1}{b_B} \right) + \frac{H_{\text{open}}^2}{a_{\text{open}} + 1}. \quad (9)$$

Three observations fix its scope.

- (a) The bound is in the heights $H_B = \max_{k \in B} |M(k) - M(m)|$. A bound $H_B \leq C b_B^{1/2+\varepsilon}$ for all blocks is equivalent to $M(x) \ll x^{1/2+\varepsilon}$, that is to the Riemann Hypothesis; and [1, §12, the equivalence (iv)] states directly that the native block budget through N is $O(N^\varepsilon)$ if and only if the Riemann Hypothesis holds. The bound cannot be used with the crude height $\max_{k \leq N} |M(k)|$: that would require $\max_{k \leq N} |M(k)| \ll N^{1/4+\eta/2} \sqrt{R(m)}$, which in the regime $R(m) \ll m^\varepsilon$ of the target is false, since $M(x) = \Omega(\sqrt{x})$. It has to be used block by block with H_B of order $b_B^{1/2+\varepsilon}$ and block lengths of order a_B , which is RH together with closure-time control.
- (b) The history term in (9) is not small on its own: with the elementary bound $|m_1(x)| \leq 1^1$ one gets only $|2M(m)h(\mu \mathbf{1}_{(m,N]})| \leq 4|M(m)| \leq 4\sqrt{mR(m)}$, which exceeds $N^\eta R(m)$ unless $R(m) \gg N^{1/2-2\eta}$. It is small only in combination with F (the combination is (2)) or if $\sum_{m < n \leq N} \mu(n)/n$ is bounded by $N^\eta R(m)/|M(m)|$, which is again a statement of RH strength.
- (c) The triangular path of [1, §9, the limit $E_a^\Delta/a \rightarrow 2 + 2 \log \frac{3}{2} - 4 \log 2$] shows that the block geometry alone permits $E_B \asymp a_B$. No bound that uses only location, length and return can give subpower growth; the arithmetic input must be a bound on the heights.

5.2 The whole signed cofactor flow

Proved (atlas entry M44, verified here). Let p be a prime, $p \leq N < p^2$, $L = \lfloor N/p \rfloor$, $x = \mu \mathbf{1}_{[1,L]}$. Then $L < p$, so for $i \leq L$ and $j \geq 1$ one has $\max(i, pj) = pj$, and

$$E(x - U_p x) = E(x) + E(U_p x) - 2\langle x, U_p x \rangle = \left(1 + \frac{1}{p}\right) R(L) - \frac{2}{p} M(L) m_1(L). \quad (10)$$

The hypotheses $p \leq N < p^2$ are exactly what makes $\max(i, pj) = pj$; without $L < p$ the cross term is not $M(L)m_1(L)/p$. From $|m_1(L)| \leq 1$ (footnote in §5.1) and $M(L)^2 \leq LR(L)$ (Lemma 2.1),

$$\left| E(x - U_p x) - \left(1 + \frac{1}{p}\right) R(L) \right| \leq \frac{2\sqrt{LR(L)}}{p},$$

and this implies the supplied two-sided bound $(1 \mp \sqrt{L}/p)R(L)$ exactly when $R(L) \geq 4L/(\sqrt{L} - 1)^2$. For smaller $R(L)$ the supplied bound needs $|2M(L)m_1(L)| \leq (\sqrt{L} - 1)R(L)$ separately; that inequality holds for $2 \leq L \leq 10^6$ with ratio at most 0.55 (Section 7) and is not implied by (10) alone. Under the Riemann Hypothesis it holds for all large L since $|M(L)m_1(L)| \ll L^\varepsilon$.

Transfer. $W = \sum_{m < p \leq N} U_p(\mu \mathbf{1}_{[1, L_p]})$, so $\sum_p (\mu \mathbf{1}_{[1, L_p]} - U_p \mu \mathbf{1}_{[1, L_p]}) = \sum_p \mu \mathbf{1}_{[1, L_p]} - W$. The vector $\sum_p \mu \mathbf{1}_{[1, L_p]}$ has coefficient $\mu(n)(\pi(N/n) - \pi(m))$ at $n \leq m$; it is not $a = \mu \mathbf{1}_{[1, m]} + u_0$, and the difference is of size $\mu(n)N/(n \log N)$ per coefficient. The incoming packet $a - W$ is therefore not a sum of single-prime flows $x_p - U_p x_p$, and (10) cannot be summed over p to bound $E(W)$, $\langle a, W \rangle$ or Δ_N . What (10) does control is the diagonal $E(v_p) = R(L_p)/p$ of Corollary 4.2, which is at most $(\log 2 + o(1))R^*(m)$ in total, while $E(W)$ itself is at least $m/(25 \log^2 m)$.

¹From $\sum_{n \leq x} \mu(n) \lfloor x/n \rfloor = 1: x m_1(x) = 1 + \sum_{2 \leq n \leq x} \mu(n) \{x/n\}$, and the last sum has at most $\lfloor x \rfloor - 1$ terms of absolute value less than 1, so $|x m_1(x)| \leq \lfloor x \rfloor$.

5.3 Joint covariance of the prime copies

Proved (atlas entry M41, verified). The Gram matrix $K_{pq} = \sum_{i \leq N/p} \sum_{j \leq N/q} \mu(i)\mu(j) / \max(pi, qj)$ is $\langle v_p, v_q \rangle$, and $E(W) = \sum_{p,q} K_{pq}$. Copies in a common quotient cell $L_p = L_q = L$ have $K_{pq} = \sum_{i,j \leq L} \mu(i)\mu(j) / \max(pi, qj)$, an exact expression; neighbouring differences inside a cell are orthogonal to later coordinates because they have zero mass.

Transfer. By Corollary 4.2 the off-diagonal part of the Gram matrix is at least $(1-o(1))m/(24 \log^2 m) - (\log 2 + o(1))R^*(m)$: the common component is not a correction to the diagonal, it is all of $E(W)$. The cells also cancel against one another. The single cell $L = 1$, the primes in $(N/2, N]$, has energy $\geq (1+o(1))N/(4 \log^2 N)$ by Lemma 2.1 and the prime number theorem, which at $N = 10^6$ is about 1.3×10^3 against the computed $E(W) = 104.6$ (Section 7). An exact within-cell Gram matrix therefore has to be combined with the signed cross-cell terms before any norm is taken; keeping the common component “explicitly included” is necessary and is not an estimate.

5.4 Record reductions

Remaining hypothesis (atlas entries M15, M16, M38). M15 shows every strict R -record is a first passage of $|M|$; M16 shows that long records cost at most a bounded amount in the square-root envelope and reduces the needed estimate to a bound $M(N)^2/N \ll_\varepsilon N^\varepsilon (1 + S(N^{1/6}))^5$ on a thinned set of records; M38 is the induction reproduced in Theorem 3.1. Each reduces the set of N at which a bound is required and leaves the strength of the bound unchanged: a bound on $|M(N)|/\sqrt{N}$ at its first passages is a bound on $\sup_{x \leq N} |M(x)|/\sqrt{x}$, and Theorem 3.2 is the quantitative form of this for H_{record} itself. The atlas labels each of these estimates open, and nothing in the present note changes that label.

6 The unsupported step

By Proposition 2.3, H_{record} is the statement that at every large strict record

$$\sum_{k=m}^{N-1} \frac{M(k)^2}{k(k+1)} + \frac{M(N)^2}{N} - \frac{M(m)^2}{m} \leq A_\eta N^\eta R(m) : \quad (11)$$

the mean square of $M(k)/\sqrt{k}$ over $(m, N]$ in the measure dk/k , at the records of R , is at most a subpower multiple of the same quantity over $[1, m]$. Every supplied result is either an exact identity that preserves (11) (the debit identity, the block decomposition, the cofactor-flow identity, the Gram matrix) or a bound of the form $\text{energy} \leq \text{height}^2 \times \text{weight}$ in which the heights are values of M . The unsupported step is the bound on those heights, equivalently on the left side of (11), below the unconditional level $N \exp(-c(\log N)^{3/5}(\log \log N)^{-1/5})$ of Theorem 3.2(c). By Theorem 3.2(a) every improvement of the exponent below 1 is a zero-free half-plane, and the exponent $\eta \rightarrow 0$ is the Riemann Hypothesis.

In the prime-side form, the same step reads: show that the signed cross term $2\langle a, W \rangle$ cancels $E(u_0) + 2M(m)h(u_0) + E(W)$ to within $A_\eta N^\eta R(m)$, while each of these three quantities is at least $m/(25 \log^2 m)$ (Theorem 4.1); that is a relative precision of order $N^{\eta-1/2}$. By Proposition 2.4(v) the cancellation is termwise, $A(k)^2 + S(k)^2 - 2A(k)S(k) = M(k)^2$, so no estimate of A or of S separately, and no regrouping of the prime copies among themselves, can replace an estimate of M on $(m, N]$. The supplied bounds are compatible with Δ_N as large as the unconditional level, and none of them addresses the cancellation.

The strongest statements established here toward the target are Theorem 3.2 (the exact strength of $H_{\text{record}}(\theta)$ for each θ), Theorem 4.1 with Corollaries 4.2 and 4.3 (the sign and the size of the cross term, and the impossibility of a term-by-term proof in the regime of the conclusion), and the identity (10) with its exact hypotheses. They are what the target rules out and what any proof has to go through; they do not reduce its strength.

7 Verification

Computation 7.1 (Identities and orders of magnitude). All quantities were computed in double precision from a sieve of μ and of the largest prime factor; the identity (1) is satisfied to 10^{-13} .

	$N = 10^6, m = 1000$	$N = 10^7, m = 3162$
$R(N)$	1.7082	1.8365
$R(m)$	1.4594	1.5409
Δ_N	0.2488	0.2957
$E(u_0)$	101.41	529.47
$2M(m)h(u_0)$	0.285	1.446
$E(W)$	104.58	541.01
$-2\langle a, W \rangle$	-206.03	-1071.63
$\sum_p E(v_p)$	0.899	0.927
$(\log 2)R^*(m)$	1.205	1.205
$m/(24 \log^2 m)$	0.87	2.03
$N/\log^4 N$	27.5	148.2
strict records $\leq N$	36	227
last record N' ; $\Delta_{N'}/R(\lfloor \sqrt{N'} \rfloor)$	355733; 0.229	6481601; 0.274

At $N = 10^6$: $S(k) = \pi(k) - \pi(m)$ for $m < k < 2(m+1)$ was checked exactly; $S(k) \log^2 k/k = -1.02, -1.10, -1.22, -1.25$ at $k = N/8, N/4, N/2, N$; the energy of the single cell $L = 1$ is 1681.8. The identity (10) was checked to 10^{-12} at $(N, p) = (10^5, 1009), (10^5, 317), (10^6, 1009)$, and $|2M(L)m_1(L)| \leq (\sqrt{L} - 1)R(L)$ holds for $2 \leq L \leq 10^6$ with maximal ratio 0.546 at $L = 3$; also $|m_1(L)| \leq \frac{1}{2}$ for $2 \leq L \leq 10^6$.

References

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- [4] A. Ivić, *The Riemann zeta-function*, Wiley, 1985, Chapter 12 (the prime number theorem with error term and the corresponding bounds for $M(x)$, Theorem 12.7).